



Integrating fuzzy metrics and negation operator in FCM algorithm via genetic algorithm for MRI image segmentation

Fatih Kutlu¹ · İbrahim Ayaz² · Harish Garg³

Received: 13 January 2024 / Accepted: 9 May 2024 / Published online: 4 June 2024
© The Author(s) 2024

Abstract

In this study, we redefine FCM algorithm by integrating fuzzy set theory, fuzzy metrics, and Sugeno negation principles. This innovative approach overcomes the limitations inherent in conventional machine learning models, especially in situations characterized by uncertainty, noise, and ambiguity. Our model utilizes the membership degrees from fuzzy set theory, and transforms the concept of proximity defined by fuzzy metrics into a minimization problem. This transformation is achieved using a linguistic negation operator, which is crucial for optimizing FCM algorithm's objective function. A significant innovation in our research is the use of GA for optimizing parameters within the contexts of fuzzy metrics and Sugeno negation. The precise optimization capabilities of GA greatly enhance the sensitivity and adaptability of FCM algorithm, thereby improving overall performance. By leveraging the meticulous parameter adjustments provided by GA, our approach has shown superior results in practical applications, such as brain MRI image segmentation, surpassing traditional methods. Experimental results highlight the considerable enhancements our proposed FCM algorithms bring over existing methods across various performance metrics. In conclusion, this study makes a valuable addition to the field of fuzzy-based machine learning methodologies. It combines the optimization strength of GA with the flexible classification capabilities of fuzzy logic. The integration of Sugeno negation and fuzzy metrics not only improves the accuracy and precision of FCM algorithm but also provides significant benefits in handling complex and ambiguous datasets. This research signifies a major advance in machine learning and fuzzy logic, setting the stage for future applications and studies.

Keywords Fuzzy C-means · Fuzzy metrics · Sugeno negation · Genetic algorithms · Image segmentation

1 Introduction

The concept of fuzzy set theory, introduced by Zadeh [1], revolutionized the way we approach problems characterized by uncertainty and ambiguity. Unlike classical set

theory, which confines elements to a binary membership of either belonging to or not belonging to a set, fuzzy set theory introduces the idea of partial membership. This allows for a more nuanced representation of real-world phenomena, where the boundaries between categories are not always clear-cut. By quantifying the degree of membership of elements to sets, fuzzy set theory provides a powerful mathematical framework to model the vagueness inherent in many complex systems, ranging from decision-making processes to pattern recognition and beyond. Recently, the incorporation of fuzzy concepts into machine learning methods has significantly increased, offering strong solutions to complicated issues [2]. Traditional machine learning algorithms often operate under the assumption of crisp, deterministic logic, where data points are assigned to distinct categories or classes. However, this deterministic approach may not be suitable for many complex scenarios where data points do not clearly belong to a single category but rather have degrees of membership

✉ Fatih Kutlu
fatihkutlu@yyu.edu.tr

İbrahim Ayaz
iayaz@beu.edu.tr

Harish Garg
harishg58iitr@gmail.com; harish.garg@thapar.edu

¹ Department of Artificial Intelligence and Robotics, Van Yüzüncü Yıl University, Van, Turkey

² Department of Computer Technologies, Bitlis Eren University, Bitlis, Turkey

³ Department of Mathematics, Thapar Institute of Engineering and Technology (Deemed University), Patiala, Punjab 147004, India

in multiple categories. The concept of degrees of membership, a cornerstone of fuzzy set theory, provides a more nuanced and flexible approach to categorization, making it a valuable addition to the machine learning toolkit for handling real-world problems characterized by imprecision, noise, and ambiguity.

Among fuzzy machine learning methods, FCM algorithm, initially defined by Dunn [3] and later improved by Bezdek [4], has gained widespread acceptance as the most prevalent technique in the arena of clustering. Unlike the K-means algorithm, which rigidly assigns each data point to a single cluster, FCM algorithm allows data points to have varying degrees of membership across multiple clusters, offering a more flexible and robust solution to real-world clustering problems [5].

Recent advancements in machine learning have highlighted FCM algorithms versatility and effectiveness of FCM algorithms across a range of complex applications. This algorithm has been pivotal in enhancing the performance of various machine learning algorithms through its application in diverse areas, such as pattern recognition, image segmentation, brain tumor classification, and even in fields extending to natural language processing and regression analysis. Starting with the integration of FCM in automatic clustering for enhanced pattern recognition through GA [6], the evolution of its applications spans sophisticated methodologies like patch-based fuzzy local similarity c-means for image segmentation [7] and extends to the classification of brain tumors using super resolution and convolutional neural networks in tandem with FCM [8]. The algorithm has also been utilized in real-time online pattern recognition using array sensors [9], and in a novel brain MRI image segmentation method that leverages an improved multi-view FCM clustering algorithm [10].

Furthermore, the adaptability of FCM is showcased through its integration with the whale optimization algorithm for innovative image segmentation approaches [11] and the construction of TSK fuzzy regression models utilizing FCM for enhanced data analysis precision [12]. Notably, FCM has also been applied in the automatic recognition of face masks using BPNN [13], in anti-noise image segmentation methods to improve reliability in noisy environments [14], and in the extraction of topics from textual data collections, demonstrating its versatility in handling unstructured data [15]. In more specific applications, FCM has contributed to the development of safety warning models for coal faces using fuzzy clustering and neural networks [16] and the enhancement of anomaly detection in databases through a FCM-based isolation forest method [17]. Lastly, the forecasting of water quality, leveraging data decomposition and deep learning neural networks in conjunction with fuzzy clustering, highlights

the algorithm's capability in environmental applications [18].

These studies underscore FCM algorithm's significant role in advancing the field of machine learning through its flexibility and effectiveness in addressing and solving multifaceted problems across various disciplines. Fuzzy clustering methodologies can be broadly divided into two distinct paradigms: one predicated on the concept of fuzzy relations and the other hinging on the strategic utilization of the objective function [19]. Fuzzy relations delve into the intricate structural interconnections among entities, characterized by degrees of similarity or dissimilarity, thereby offering a nuanced perspective on their relationships. On the other hand, objective function-based algorithms ingeniously transform the complex issue of clustering into an optimization problem. The degree of homogeneity within the cluster is meticulously gauged through the objective function, and the optimal partitioning is achieved by minimizing this objective function. These methods necessitate a clear understanding of the number of clusters and the specific attributes of the cluster prototypes and exhibit a high degree of sensitivity to the initial values in the clustering process [20].

The transformative capability of fuzzy set theory to reconceptualize classical sets establishes a broader framework for understanding mathematical constructs. This paradigm, having yielded a panoply of comprehensive concepts, positions itself as a potentially superior alternative to conventional methodologies. A case in point is the notion of distance, which, under the lens of fuzzy metrics, obtains a novel interpretation [21–26]. The integration of the fuzzified interpretation of distance, as offered by fuzzy metrics, could significantly augment FCM algorithm. Defined by a degree of closeness, the fuzzy metric could engender a robust clustering mechanism adaptable to intricate data scenarios, thereby yielding a versatile model proficient at handling heterogeneous data structures and noise in image segmentation [27–29].

The aim of this study is to redefine FCM algorithm using fuzzy metrics and a negation operator. Since the fuzzy metric is defined based on a degree of closeness, we use a linguistic negation operator to express this degree as a degree of distance, thus converting it into a minimization problem of the objective function, including the original FCM. The use of this negation operator actually involves a fuzzification process when converting the concept of proximity to distance. This allows us to obtain a more precise calculation mechanism with a greater number of meaningful parameters. Both the parameters encountered in the fuzzy metric and the parameters encountered in the negation operator are optimized using GA. We have tested this innovative approach on brain MRI image segmentation, where it demonstrated practical effectiveness.

Therefore, this paper aims to explore the potential of redefining FCM algorithm within the boundaries of fuzzy set theory and fuzzy metrics. By examining the theoretical foundations, real-world applications, and possible challenges of this paradigm shift, it aims to promote a more detailed understanding of clustering algorithms and accelerate further progress in the field. In this study, we illustrate how FCM algorithm categorizes data points within a multidimensional space and determines the centers of these groups based on the geometric locations of data points. By evaluating the 'distance' of each data point to the group centers using fuzzy metrics, we reveal the algorithm's robustness from a geometric perspective. This geometric framing not only enhances our understanding of the algorithm's mechanics but also underscores its applicability in handling complex data structures inherent in machine learning tasks. To better elucidate the contributions of this research, we succinctly list the significant advancements made by our study below:

1. *Innovative redefinition of FCM algorithm:* We introduce a novel approach by integrating fuzzy set theory, fuzzy metrics, and Sugeno negation principles into FCM algorithm, significantly enhancing its performance, especially in handling data characterized by uncertainty, noise, and ambiguity.
2. *Enhanced clustering mechanism via fuzzy logic:* By employing fuzzy logic, including fuzzy metrics for proximity assessment and Sugeno negation for distance interpretation, we provide a more flexible and accurate clustering mechanism. This advancement allows for more nuanced data segmentation, which is particularly beneficial for medical imaging like MRI.
3. *Parameter optimization with GA:* Our research employs GA for the optimization of the parameters within the fuzzy metrics and the negation operator. This optimization improves the algorithm's sensitivity and adaptability, leading to superior segmentation outcomes.
4. *Superior performance in MRI image segmentation:* Through rigorous experimentation, our modified FCM algorithm demonstrates significant improvements over traditional methods in the segmentation of brain MRI images. The enhancements are evident in accuracy, precision, recall, and overall segmentation quality, underscoring the potential of our methodology to refine diagnostic processes.
5. *Framework for future fuzzy-based machine learning models:* Our study not only advances the field of machine learning in the context of MRI image segmentation but also sets a foundation for future research. By highlighting the effectiveness of

integrating fuzzy logic with GA, we pave the way for exploring other complex data analysis tasks.

These contributions underline the significance of our work, advancing the capabilities of machine learning models to address and solve multifaceted problems across various disciplines, particularly in medical imaging analysis.

The paper is divided into several sections: Sect. 2 lays out basic concepts related to fuzzy sets and fuzzy metrics, setting the groundwork for our redefined FCM algorithm. Section 3 introduces our proposed FCM algorithm and provides information about GA used for optimization. The final section showcases experimental results, demonstrating the real-world application of our model in brain MRI image segmentation. To ensure clarity and ease of understanding, the abbreviations used throughout this manuscript are listed below (Table 1):

2 Related works

The segmentation of brain MRI images is a pivotal component in medical image analysis, necessitating the precise delineation of brain tissues for accurate diagnosis, treatment planning, and the monitoring of brain conditions. Over the years, FCM algorithm and its enhancements have been instrumental in pushing the boundaries of segmentation accuracy and efficiency. This section encapsulates significant strides made in brain MRI segmentation, with a special focus on innovations in FCM techniques.

The foundational contribution by Pham and Prince [30] laid the groundwork for FCM-based segmentation methods, addressing the challenges posed by noise and intensity inhomogeneities in MRI images. Following this, Siyal and Yu [31] demonstrated how modifications to the standard FCM could enhance segmentation in the presence of

Table 1 Abbreviations

Abbreviation	Full term
FCM	Fuzzy C-means
GA	Genetic algorithm
MRI	Magnetic resonance imaging
SE	Sensitivity
SP	Specificity
PR	Precision
F1	F1 score
ACC	Accuracy
IoU	Intersection over union
N/A	Not applicable

intensity inhomogeneities. Singh and Bala [32] further advanced this field by proposing a DCT-based local and non-local FCM algorithm that balanced noise reduction with the preservation of image details.

Bai et al. [33] introduced an improved probabilistic FCM method that, through the application of a similarity measure, effectively mitigated the “cluster-size sensitivity” problem while bolstering resistance to noisy images. That same year, Huang et al. [34] merged FCM clustering with rough set theory to achieve superior segmentation outcomes for fuzzy boundary regions. Liu et al. [35] proposed a novel FCM approach that utilized multiple-surface approximation and interval memberships for bias correction and segmentation of brain MRI, marking a significant methodological advance.

The work of Valsalan et al. [36] presented a knowledge-based FCM method that significantly expedited brain tissue segmentation from MRI scans by leveraging a CUDA-enabled GPU machine. Tavakoli-Zaniani et al. [37] developed a modified FCM algorithm based on double estimation, which proved effective in segmenting brain structures from noisy MR images, demonstrating enhanced robustness against noise.

Integrating evolutionary algorithms for learning model optimization, our study introduces a distinct synthesis of FCM algorithm enhancements, fuzzy set theory, and GA optimization, specifically tailored for MRI image segmentation. Unlike the focused applications seen in [38, 39], which apply FCM and GA in supervisory control and disease diagnosis, respectively, our research pioneers the use of these methodologies to significantly enhance segmentation accuracy within the complex domain of medical imaging. Neves-Jr. et al.’s exploration into the chemical industry processes and Chen et al.’s advancement in Traditional Chinese Medicine diagnosis demonstrate the versatility of combining FCM with GA; however, our approach optimizes this integration for the nuanced challenges inherent in MRI image segmentation. This optimization includes a detailed focus on fuzzy metrics and Sugeno negation principles, key to addressing noise and ambiguity in segmentation tasks. Furthermore, our methodology differentiates itself from [40], who investigated the efficiency of GA in learning FCM through metaheuristic methods, by not only utilizing GA for supplementary optimization but embedding it as a core component of redefining FCM algorithm itself. The comparison with simulated annealing underscores the potential of evolutionary algorithms in refining learning models, yet our implementation showcases a novel application in medical imaging that leverages GA’s robust optimization capabilities more fully. Additionally, Wang et al. [41] highlighted GA’s utility in optimizing cluster results for signal sorting, illustrating GA’s broad applicability in

solving complex clustering challenges. Our study, however, extends these concepts to the medical imaging domain, demonstrating a notable enhancement in MRI image segmentation. Our experimental findings on the BraTS2018 dataset affirm the efficacy of our approach, revealing significant improvements in accuracy, precision, and recall rates compared to conventional methods.

Building on the seminal works in the field, our research marks a notable advancement in the utilization of evolutionary algorithms for the optimization of learning models, particularly within medical imaging analysis. We present a transformative enhancement of FCM algorithm through an intricate integration of GA for parameter optimization, alongside the application of fuzzy set theory, fuzzy metrics, and the Sugeno negation operator. This multifaceted approach not only effectively navigates the challenges posed by data ambiguity, noise, and imprecision but also fosters a more flexible and refined strategy for addressing segmentation dilemmas.

The innovative melding of these technologies within our study not only illuminates the superior effectiveness of an evolved FCM algorithm in medical image segmentation but also signifies a substantial leap forward in the field. By melding the optimization strength of GA with the adaptive classification capabilities of fuzzy logic, we unveil a robust and nuanced solution to the complexities of medical imaging. This research endeavors to significantly enrich the corpus of FCM-based segmentation methodologies, establishing a foundational framework for future explorations and advancements in medical imaging analysis.

3 Fuzzy sets and fuzzy metric

Definition 3.1 [1] A fuzzy set A on non-empty set X is a set of ordered pair $A = \{(x, \mu_A(x)) : x \in X\}$ where $\mu_A : X \rightarrow I = [0, 1]$. For each $\forall x \in X$, the value $\mu_A(x)$ is called as membership degree of x to A and the function μ_A is called membership function of A . Let denote set of all fuzzy sets on X by FS^X .

Each classical (crisp) set can be interpreted as a fuzzy set with its characteristic function, suggesting that fuzzy set theory provides a more expansive perspective than classical set theory. By re-examining various mathematical concepts through the lens of fuzzy set theory, we can generate new, wider, and more practical concepts that surpass traditional methods. The concept of distance is a particularly noteworthy example of these notions. In fuzzy set theory, distance, often explored through the notion of a metric, has many generalizations, with the fuzzy metric concept being a key example. The measure of proximity or closeness, used in the fuzzy metric concept, indicates the

distance between points. Before providing the definition of fuzzy metrics, let us first establish the definitions of several key concepts that will be employed within the proposed FCM algorithm.

Definition 3.2 [42] If $N : [0,1] \rightarrow [0,1]$ function satisfies the following conditions, N is called a negation.

1. $N(0) = 1, N(1) = 0$,
2. N is a non-increasing function, i.e., $x \leq y \Rightarrow N(x) \geq N(y), x, y \in [0,1]$.

If a negation is monotonously decreasing, i.e., $x < y \Rightarrow N(x) > N(y)$ for $x, y \in [0,1]$ and continuous, it is called a strict negation. If a strict negation is an involution, i.e., $N(N(x)) = x$ for $x \in [0,1]$, it is called a strong negation.

Definition 3.3 [42] If the binary operator $T : [0,1] \times [0,1] \rightarrow [0,1]$ satisfied the following conditions for all $a, b, c, d \in [0,1]$ then it is called a triangular norm (briefly t-norm).

1. Bounded Condition: $T(1, a) = a$
2. Monotonicity: $a \leq c$ ve $b \leq d \Rightarrow T(a, b) \leq T(c, d)$,
3. Commutativity: $T(a, b) = T(b, a)$,
4. Associativity: $T(a, T(b, c)) = T(T(a, b), c)$

As can be understood from these properties, the t-norm is a monotonously non-decreasing function and provides $T(a, 0) = 0$. Examples of the most used t-norms are minimum, algebraic multiplication, and Lukasiewicz t-norms defined by $T(a, b) = \max\{0, a + b - 1\}$.

The concept of fuzzy metric, initially defined by Kramosil and Michalek [21] and later revisited by George and Veeramani [22] to derive the Hausdorff topology, holds a significant place. This definition focuses not on the concrete distance between two points but rather on the fuzzy grading that gives rise to such a distance.

Definition 3.4 [22] The 3-tuple $(X, M, *)$ is said to be a fuzzy metric space if X is an arbitrary set, $*$ is a continuous t-norm and M is a fuzzy set on $X^2 \times (0, \infty)$ satisfying the following conditions:

1. $M(x, y, t) > 0$
2. $M(x, y, t) = 1$ if and only if $x = y$
3. $M(x, y, t) = M(y, x, t)$
4. $M(x, y, t) * M(y, z, s) \leq M(x, z, t + s)$
5. $M(x, y, \cdot) : (0, \infty) \rightarrow [0, 1]$ is continuous.

where $x, y, z \in X$ and $t, s > 0$. In this case, $M(x, y, t)$ is referred to as the proximity of x and y with regard to t . In other terms, it refers to the truth of the statement “ x is as close to y as t ”. In [22, 25], several methods are defined for deriving fuzzy metrics from traditional metrics. Following the definition of a fuzzy metric space, it is crucial to underscore the significance of each condition detailed within this

framework. At its core, a fuzzy metric space extends the traditional notion of a metric space by incorporating the concept of ‘fuzziness’, reflecting the real-world ambiguity and gradual variation in distances between points:

1. Nonzero condition: This ensures that the fuzzy distance between any two points is always positive, except when the points coincide, enhancing the model’s robustness by accounting for the subtle gradations in proximity that classical metrics might overlook.
2. Identity of indiscernible: By stipulating that the fuzzy distance equals one if and only if the points are identical, this condition aligns with the intuitive understanding of distance, reinforcing the metric’s relevance to real-world scenarios where exact matches are distinct from close resemblances.
3. Symmetry: The requirement that the fuzzy distance from point X to Y is the same as from Y to X mirrors the physical world’s symmetry in distances, ensuring the metric’s applicability across different domains without bias.
4. Triangle inequality: A cornerstone of metric spaces, this condition is adapted to the fuzzy context to maintain the essential logic of distances within a more flexible framework. It ensures that the direct distance between two points is always less than or equal to the distance via a third point, preserving the intuitive notion of distance despite the introduction of fuzziness.
5. Continuity: The continuity of the fuzzy metric with respect to time guarantees that small changes in the temporal parameter do not lead to abrupt changes in perceived closeness, reflecting the gradual shifts that characterize real-world phenomena.

Each of these conditions plays a pivotal role in defining a coherent and applicable metric space that captures the essence of fuzziness, enabling the precise modeling of systems and processes fraught with uncertainty and imprecision. The integration of fuzzy metric spaces into our study lays the groundwork for leveraging this mathematical construct in the enhancement of FCM algorithm for MRI image segmentation. This approach not only acknowledges but also embraces the inherent ambiguity in medical imaging, aiming for a segmentation method that is both nuanced and highly adaptable to the complex, layered nature of brain MRI data. The innovative aspect of this notion lies in its ability to develop a more universal and adaptable distance measurement while preserving the essential topological characteristics typically associated with metric spaces. This is done by considering fuzzy metrics that are not derivable from classical metrics. The concept of fuzzy metric fuses the established principles of classical topology with the computational advantages of degree of closeness. Consequently, this allows for problems to be addressed with a more comprehensive,

linguistically nuanced approach while maintaining their inherent topological and geometric attributes.

4 Proposed FCM algorithm using fuzzy metrics and negation operator through GA

FCM algorithm enables the division of images based on the similarities and differences of grayscale pixels. In a marked departure from prior methodologies, FCM acknowledges that entities can belong to multiple clusters. The fundamental concept of fuzzy logic posits that each data item has a degree of membership lying between $[0,1]$, thereby creating distinctive clusters for each membership [4]. The aggregate of membership values spanning all classes for a data point must equal to 1, i.e., $u_{ij} \geq 0$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, c$:

$$\sum_{j=1}^c u_{ij} = 1 \quad (1)$$

where u_{ij} is the membership value of the i -th data to the j -th cluster. The degree of belonging to the cluster that the object is closest to will be greater than the degree of belonging to other clusters. The objective function is given in Eq. 2.

$$J(u, v) = \sum_{i=1}^N \sum_{j=1}^C u_{ij}^m \|x_i - v_j\|^2, 1 \leq m < \infty \quad (2)$$

Equation 2 consists of the following variables: X represents the data as a set $\{x_1, x_2, \dots, x_n\}$; C denotes the number of clusters in X ; m is the weighting exponent, satisfying $m \geq 1$; v_j corresponds to the center of the j -th cluster; u_{ij} represents the membership value of x_i to the j -th cluster; $\|\cdot\|$ denotes a norm defined on the vector space $\mathbb{R}^N$. FCM algorithm prototype can be obtained by minimizing the objective function $J(u, v)$ for each value of C . Specifically, this involves taking the partial derivative of $J(u, v)$ with respect to v_j and setting it to zero. The cluster centers and membership values are calculated using Eq. 3 and Eq. 4, respectively:

$$v_i = \frac{\sum_{k=1}^N (u_{ik})^m x_k}{\sum_{k=1}^N (u_{ik})^m} \quad (3)$$

$$u_{ij} = \frac{1}{\left[\sum_{k=1}^c \left(\frac{\|x_i - v_j\|}{\|x_i - v_k\|} \right)^{\frac{2}{m-1}} \right]} \quad (4)$$

Following are the stages taken by FCM algorithm (Table 2):

We will present FCM algorithm in the context of our investigation using a novel method. The primary

distinction is that the traditional distance measures will be replaced with a fuzzy metric-based distance measure. The fuzzy distances calculated using the fuzzy metric are changed from the degree of proximity to the degree of distance using a negation in order to produce a fuzzy metric-based distance measure. At this point, negation operators of the Yager or Sugeno types are primarily employed. As a result, by considering the degree of proximity rather than language, the epistemic meaning of distance is provided. One of the novel techniques of this work is the application of the proximity-based distance measure in FCM algorithm. At this point, it is important to convert the fuzzy metric's measurement of closeness into a measurement of distance from the dual. This transition is accomplished by using strong negations. In this work, the following fuzzy metrics and negation are used:

- i. Let (X, d) be any classical metric space (e.g., $(\mathbb{R}, |\cdot|)$). Define $a * b = a.b$ and

$$M_1(x, y, t) = \frac{1}{e^{d(x,y)/t}}$$

for all $x, y \in X$ and $t > 0$. Then $(X, M_1, *)$ is a fuzzy metric [22].

- ii. Let (X, d) be any classical metric space (e.g., $(\mathbb{R}, |\cdot|)$). Define $a * b = a.b$ and

$$M_2(x, y, t) = \frac{\alpha_1 t^{\alpha_3}}{\alpha_1 t^{\alpha_3} + \alpha_2 d(x, y)}$$

for all $x, y \in X$ and $\alpha_1, \alpha_2, \alpha_3, t > 0$. Then $(X, M_2, *)$ is a fuzzy metric [25].

- iii. Let $X = \mathbb{R}^+$ and define $a * b = a.b$ and

$$M_3(x, y, t) = \left(\frac{(\min\{x, y\})^{\alpha_1} + t}{(\max\{x, y\})^{\alpha_1} + t} \right)^{\alpha_2}$$

for all $x, y \in X$ and $\beta_1, \beta_2, t > 0$. Then $(X, M_3, *)$ is a fuzzy metric [25].

To interpret these closeness's into distances from the dual, employ Sugeno Negation defined as follow:

$$N_S(x) = \frac{1 - x}{1 + \lambda x}$$

for $\lambda > -1$ and $x \in [0, 1]$ [42].

The objective function in this proposed algorithm is defined as follows:

$$J(u, v) = \sum_{i=1}^N \sum_{j=1}^C u_{ij}^m d_{\text{fuzzy}}(x_i, v_j)^2, 1 \leq m < \infty \quad (5)$$

where $d_{\text{fuzzy}}(x_i, v_j) = N(M(x_i, v_j, t))$ for N is a strong fuzzy negation and M is a fuzzy metric. The structure of this novel clustering algorithm, as defined by the fuzzy metric, mirrors that of the traditional FCM algorithm. By

Table 2 FCM algorithm

Begin
Fix $c, 2 < c < n$;
Fix $\varepsilon > 0$;
Fix $\text{maxIteration} \in \mathbb{N}$;
Choose type of inner product norm metric;
Fix $m, m \geq 1$;
Fix initial matrix $U^0 \in M_{fc}$ of u_{ij} .
For $k = 1$ to maxIteration do
Calculate clusters centers V using Eq 7.
Update membership matrix ($U_{k+1} = [u_{ij}^{(k+1)}]$) with Eq 8.
Calculate $\ U_{k+1} - U_k\ $ for any matrix norm
if $\ U_{k+1} - U_k\ < \varepsilon$ then
break
Else
$U_k = U_{k+1}$
end if
end for
end

maintaining the logical framework of FCM and introducing fuzziness to the concept of distance within it, a new clustering approach has been developed. This method upholds the topological features of the traditional approach while capitalizing on the linguistic and logical advantages offered by fuzzy set theory. Within this innovative approach that employs fuzzy negation and degree of closeness, it is apparent that several variables can significantly influence the clustering results. The efficacy of the clustering process, which involves extracting necessary features from the primary image, is contingent on the choice of these parameters. Consequently, the utilization of GA for parameter optimization has led to the development of a more proficient method.

5 Explanation of programmatic model construction

To elucidate the construction of our programmatic model, this section is dedicated to detailing the parameters, simulation environment setup, and genetic parameters utilized to initialize GA which plays a pivotal role in optimizing our modified FCM algorithm.

5.1 Simulation environment setup

The simulation environment for our study was meticulously architected to facilitate rigorous testing and validation of the proposed algorithm. This computational setup was instantiated on a system equipped with an AMD Ryzen

7 CPU, an NVIDIA RTX A5000 GPU, and 128 GB of RAM, operating under Windows 11. The algorithm itself was implemented using Python 3.8, which was chosen for its robust capabilities in numerical computations and optimization routines.

5.2 Parameters definition and GA initialization

In our study, numerous parameters exist within the integrated fuzzy metrics and negation operator, necessitating optimization. These parameters, inherently multitudinous, are not merely a complication but rather a rich source of fine-tuning potential, offering a higher degree of precision in the algorithm's performance. They enable a more sophisticated modeling of the problem at hand and, therefore, more accurate segmentation in our application of MRI imaging. These parameters have been optimized using GA, demonstrating the utility of evolutionary computation methods in handling the complexity and multidimensionality of the optimization landscape within the context of our enhanced FCM algorithm. GA is a computational method that draws inspiration from the principles of genetics and natural selection, serving as a powerful optimization and search technique. It is one of the earliest population-based stochastic algorithms ever proposed, with its roots deeply embedded in the biological evolution paradigm [43–45]. GA operates through a series of processes that mimic natural evolution, including evaluation, selection, crossover (recombination), and mutation [44]. These processes are applied to a population of candidate solutions to an optimization problem, with the aim of evolving the population toward better solutions. In the context of GA, these candidate solutions are represented as vectors of equal size, often referred to as chromosomes. The chromosomes are assembled randomly to form a population. Each chromosome encodes a potential solution to the problem at hand. The fitness of each solution is evaluated using an objective function, also known as the fitness function. This function quantifies the quality of the solutions, guiding the selection process where better solutions (chromosomes with higher fitness) are given a higher chance to be selected for reproduction. During the crossover process, pairs of chromosomes exchange parts of their structure, creating new offspring that combine traits from both parents. The mutation process introduces small random changes in the offspring, ensuring diversity in the population and preventing premature convergence to sub-optimal solutions. As the algorithm progresses, generations of solutions are produced, each hopefully better than the last, as they are guided by the fitness function toward optimal or near-optimal solutions to the problem. This iterative process of selection, crossover, and mutation continues until a satisfactory solution is found or a

predefined termination condition is met. Indeed, in recent years, there has been an increasing trend toward utilizing GA for optimizing fuzzy methods. This stems from the fact that GAs are robust search algorithms based on the mechanics of natural selection and genetics, making them particularly suited for exploring complex, multidimensional spaces such as those encountered in fuzzy logic and systems. They provide a global search approach, avoiding the local minima traps that other optimization methods may fall into. Therefore, GA's evolutionary computation methodology complements the inherent complexity of fuzzy systems, allowing for a comprehensive optimization of their numerous parameters. As a result, researchers and practitioners are increasingly leveraging this combination to solve complex problems across various fields, from image segmentation to decision-making systems [46–48]. In this study, F1 score has been selected as the fitness function, and the pseudocode for GA constructed based on this selection is provided in Table 3.

In this study, we leverage GA for optimizing parameters of FCM algorithm, specifically devised for the segmentation of brain MR images. Our approach adopts an F1-Score-based fitness function, aiming to identify the optimal set of parameters that maximizes the performance of the algorithm. This optimization process notably benefits from the integration of fuzzy metrics and Sugeno negation, significantly enhancing the algorithm's effectiveness on the dataset.

The roles of fuzzy metric and Sugeno negation are crucial in this context. The fuzzy metric offers a more adaptable method for gauging the proximity between data points compared to traditional metrics, adeptly managing the data's inherent uncertainty and noise. Conversely, Sugeno negation transforms these proximity values into distances, thus effectively contributing to solving the minimization problem of FCM algorithm's objective function. This pivotal transformation bolsters the algorithm's ability to refine segmentation precision while simultaneously reducing false positives. Within the GA, the fitness function based on the F1-Score considers the distance values derived from fuzzy metrics and Sugeno negation, evaluating the efficacy of each parameter set. This synthesis guarantees a balanced performance in terms of accuracy (precision) and sensitivity (recall), thereby sharpening the segmentation outcomes' precision and minimizing the rate of false positives.

The fundamental motivation for selecting GA for this purpose stems from their exceptional capability to navigate complex, multidimensional optimization landscapes. Critical variables, including those within fuzzy metrics and the Sugeno negation, are meticulously optimized using the evolutionary computation methodology provided by GAs. Moreover, as delineated in Table 3, the F1-Score

performance metric, recognized for its efficacy in addressing class imbalance problems, is integrated into the GA's fitness function along with FCM algorithm. The fitness function creates a confusion matrix by conducting a pixel-based comparison between the U matrix generated by FCM and the ground truth image of MRI scans. This matrix forms the foundation for calculating performance based on the F1-Score metric.

Conclusively, the employment of GA in this study substantially elevates the sensitivity and adaptability of FCM algorithm. This integration facilitates more effective parameter adjustments by the algorithm in datasets characterized by complexity and uncertainty, thereby enhancing the overall segmentation results. When compared to similar studies in the literature, this approach underscores the robustness of combining F1-Score, fuzzy metrics, and Sugeno negation, illustrating how such integration can serve as a potent tool in segmentation challenges. The hyperparameters for the GA were chosen based on values commonly used in research, including a maximum iteration count of 50, population size of 30, mutation probability of 0.1, elitism rate of 0.01, and crossover probability of 0.5. These settings are designed to ensure the efficient operation of GA, facilitating the achievement of optimal or near-optimal solutions. Below is a Table 4 summarizing the optimized parameters, their symbols, descriptions, and the search ranges used in the optimization process:

This Table provides an overview of the parameters optimized using GA in this study, including their symbols, descriptions, and defined search ranges for each parameter. The selection of parameters and their search ranges has been carefully tailored to maximize the algorithm's performance and adaptability to the specific optimization problem. The incorporation of these parameters and their optimization through GA highlights the significance of combining the optimization power of GA with the flexible classification capabilities of fuzzy logic to enhance the accuracy and precision of FCM algorithm. This integration not only improves FCM algorithm's performance but also significantly benefits handling complex and ambiguous datasets, thereby marking a significant advancement in the field of machine learning and fuzzy logic.

6 Experimental results

6.1 Dataset

In this research, the comprehensive and diversified collection of MR images provided by the Brain Tumor Segmentation (BraTS2018 [49–51]) competition has been utilized as the primary source of data. The selected dataset encompasses MR images of 285 patients, classified into

Table 3 GA with F1 score as the fitness measure:

```

Start
create initial population
while iteration number < maximum number of iterations
    for each chromosome, evaluate fitness:
        start FCM
            image segmentation
            final U matrix
        end FCM
        fitness= F1 score (final u matrix, ground truth vector)
    end for
    parent select
    crossover
    mutation
    elitism
end while
output fittest chromosome
end

```

two different tumor grades: high-grade gliomas (HGG) and low-grade gliomas (LGG). This meticulously compiled dataset offers four different imaging modalities for each patient: T1-weighted images, T1 post-contrast (enhanced with contrast agent) images, T2-weighted images, and Fluid-Attenuated Inversion Recovery (FLAIR) images. This diversity allows for the distinction of different physiological and anatomical features of the tumor and surrounding tissue, thus increasing the accuracy of the segmentation process. The dataset includes corresponding ground truth data, providing a reliable benchmark for the evaluation of our segmentation results. During the data processing phase, a careful preprocessing routine has been applied to maximize the effectiveness of the segmentation algorithm. This process, aimed at ensuring a homogeneous examination of the images, includes normalization and noise reduction techniques. The normalization procedure scales the intensity values of the MR images to a standard range, minimizing variations that may arise from different scanning parameters and devices. This allows the algorithm to interpret features on the image more consistently and

thereby more clearly delineate the boundaries between tumor and healthy tissue.

In our application of MRI image segmentation, we delve into how each pixel or voxel is positioned in space to determine its affiliation with specific anatomical structures. Our proposed FCM-based algorithm performs this grouping by considering the spatial locations and intensity values of these pixels, thereby clearly delineating the geometric boundaries of structures such as tumors. The critical role of GA in optimizing these parameters significantly refines segmentation accuracy, thereby improving the geometric representation of the segmented structures. This approach not only optimizes the clustering process but also provides a vivid geometric insight into the segmentation of complex images, illustrating the potential of our method for accurately identifying and outlining critical anatomical features.

6.2 Performance measures and optimal parameters

Our approach involved conducting a pixel-based classification with FCM algorithm enhanced by three newly proposed fuzzy metrics and Sugeno negation. Each fuzzy metric was evaluated individually, offering distinct perspectives on the segmentation task. The integration of these fuzzy metrics allowed us to comprehensively explore the algorithm's performance under various conditions, contributing to a more thorough understanding of its capabilities and limitations. In addition to FCM algorithm, GA was employed to optimize hyperparameters of fuzzy metrics and Sugeno negation. GA have been proven effective in tackling high-dimensional and complex optimization problems, making them an ideal choice for our task. F1-Score, a well-established performance metric in the field of image segmentation, served as the fitness function in our GA. FCM algorithm was set with a specific number of clusters (C) equating to 6, a small epsilon value of 0.00001 to control the termination criteria, and a maximum iteration limit set at 100. These settings were carefully chosen based on preliminary tests and literature recommendations,

Table 4 Optimized parameters and their search ranges in GA optimization

Symbol	Description	Search range
m	Fuzzification parameter controlling the degree of data point spread across clusters in FCM algorithm	$1.1 \leq m \leq 4$
λ	Sugeno negation parameter affecting the shape of the Sugeno negation operator and consequently the sensitivity of distance measurement	$2 \leq \lambda \leq 4$
t	Fuzzy distance criterion reflecting the proximity between two points in fuzzy metric space	$1024 < t \leq 5000$
$\alpha_1, \alpha_2, \alpha_3$	Additional parameters for fuzzy metrics influencing the characteristics of fuzzy metrics and the distance calculations between two points	$1.1 \leq \alpha_i \leq 10, i = 1, 2, 3$

Table 5 Optimized parameters resulting from GA in FCM algorithm with fuzzy metrics and sugeno negation

Fuzzy metric	m	λ	t	α_1	α_2	α_3	Objective function (F1 score)
$M1$	1.642	3.707	4153	N/A*	N/A	N/A	86.9960%
$M2$	1.441	3.780	4.985	2.909	9.737	2.114	89.7516%
$M3$	1.158	2.321	2475	2.751	3.475	N/A	89.7831%

*Fields designated as 'N/A' represent parameters that do not have corresponding elements in the equations

ensuring a balanced trade-off between computational efficiency and result accuracy. The parameters that were optimized through the GA, including those of both FCM and fuzzy metrics, are provided in Table 5. These results demonstrate the effectiveness of the optimization process and shed light on the optimal conditions under which our proposed FCM algorithm performs best.

In the realm of brain MRI image segmentation, accurately delineating pathological structures, such as tumors, is paramount for diagnosis and therapeutic planning. However, challenges such as the intensity similarity between tumors and healthy tissues, noise, and boundary ambiguity in MRI images significantly complicate the segmentation process. Classical methodologies, including K-means and standard FCM algorithm, exhibit limited success in overcoming these obstacles. K-means algorithm, with its rigid assignment of data points to a single cluster, fails to accommodate the inherently fuzzy boundaries and varying degrees of membership characterizing biological tissues. Conversely, while standard FCM offers more flexibility in membership allocation, it remains susceptible to noise and outliers, adversely affecting segmentation quality. The proposed FCM algorithm proposed in this study, through the integration of fuzzy set theory, fuzzy metrics, and Sugeno negation principles, addresses these challenges head-on. This approach enables the algorithm to more effectively process noise, ambiguity, and the fuzzy nature of data, thereby overcoming the limitations encountered by standard FCM. Experimental evaluations conducted on the BraTS2018 dataset have demonstrated significant improvements in crucial performance metrics such as precision, accuracy, and the F1 score compared to classical methods. These enhancements are attributable to parameter optimization facilitated by GA. The optimization process, which notably enhances the algorithm's sensitivity and adaptability, has resulted in a marked increase in segmentation quality.

In conclusion, the proposed FCM algorithm, by harnessing the synergy of fuzzy logic and GA, offers innovative solutions to complex problems like brain MRI image segmentation. This study exemplifies how the integration of fuzzy logic and GA can substantially elevate the performance of machine learning models. Moreover, it anticipates the potential extension of this methodology to other

areas of medical imaging, thereby promising to improve diagnostic and therapeutic processes within the health sciences.

Pixel-based classification was executed using the hyperparameters outlined in Table 4. In order to gauge the effectiveness of our classification, we employed a confusion matrix, a powerful tool that provides a detailed breakdown of the classification results. The confusion matrix is comprised of four main components, or inF1s, as follows:

In our study, pixel-based classification was conducted using the outlined hyperparameters in Table 5. To evaluate the classification's efficacy, a confusion matrix was used. It consists of four main components:

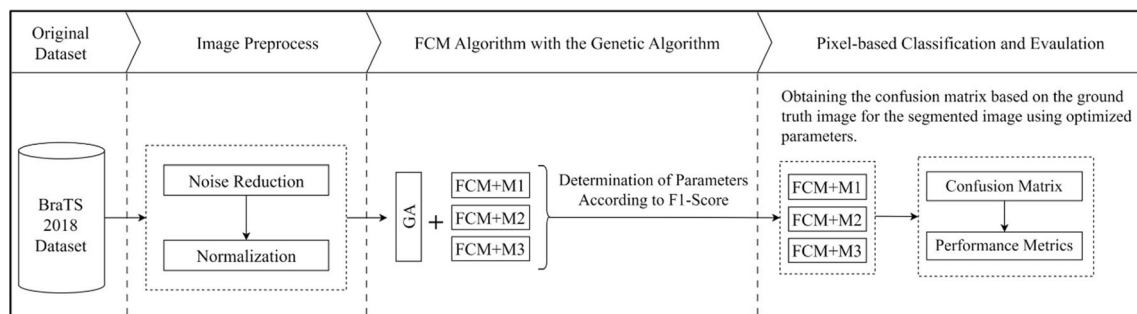
- True positives (TP): instances where our algorithm accurately segmented actual tumor areas.
- True negatives (TN): instances where the algorithm correctly recognized non-tumor areas.
- False positives (FP): instances where the algorithm incorrectly classified actual tumor areas as non-tumor.
- False negatives (FN): instances where non-tumor areas were mistakenly segmented as tumor areas by our algorithm.

It should be noted that each instance corresponds to a pixel in MRI images. Using these in F1s, we evaluated the effectiveness of our classification using performance measurement metrics. The details of these metrics can be found in Table 6. This comprehensive evaluation approach allowed us to gain a deep understanding of the algorithm's performance in brain tumor segmentation.

Following our comprehensive standardization efforts, we proceeded with executing the proposed FCM algorithms fortified with the Sugeno negation and various fuzzy metrics and standard FCM on an array of randomly selected images from the dataset. These procedures utilized optimized parameters, which were meticulously fine-tuned by GA to maximize the efficiency and effectiveness of our image segmentation endeavors. The proposed algorithms, through their intensive fuzzy metric calculations, embraced the rich complexity of the 57,600 data points in each image, fully exploiting the valuable information inherent in each pixel. The application of Sugeno negation and fuzzy metrics further enhanced the sensitivity of the proposed

Table 6 Key performance metrics definitions used for classification evaluation

Metrics	Symbol	Formulization	Evaluation focus
Sensitivity	SE	$\frac{TP}{TP+FN}$	Measures the proportion of true positives correctly identified
Specificity	SP	$\frac{TN}{TN+FP}$	Evaluates the proportion of true negatives correctly identified
Precision	PR	$\frac{TP}{TP+FP}$	Measures the proportion of identified positives that are correct
F1 score	F1	$\frac{2TP}{2TP+FP+FN}$	Harmonic mean of precision and sensitivity, indicating segmentation quality
Accuracy	ACC	$\frac{TP+TN}{TP+FP+FN+TN}$	Proportion of total correct predictions
Intersection over union	IoU	$\frac{TP}{TP+FN+FP}$	Measures overlap between predicted and ground truth segmentation

**Fig. 1** Flowchart of the brain MRI image segmentation using the proposed FCM and GA integration

FCM algorithms to variations in the pixel intensity, which is a fundamental attribute in our image segmentation task. This ultimately allowed the proposed FCMs to more accurately segment the images into tumor and non-tumor areas. In order to assess the performance of our approaches, we delved into a rigorous evaluation process that was underscored by a visual comparison of the segmented images and a detailed investigation of key performance metrics. Visually, the segmented images were juxtaposed with the corresponding ground truth images. The clarity of the segmented tumor areas in the modified FCM results as compared to those in the classical FCM was profoundly striking, demonstrating the significant improvement in the segmentation quality achieved through modifications.

Regarding performance metrics, we utilized the above-mentioned measures to assess the quality of our segmentation process. It was evident that proposed FCM algorithms consistently surpassed the performance of traditional FCM methods, signifying the enhanced effectiveness of our approach. A notable improvement was observed in the precision rate, indicative of the proposed FCM algorithm's superior capacity to accurately delineate tumor pixels. Likewise, an enhanced recall rate was witnessed, marking a substantial increase in the identification of actual tumor pixels. Further indicators of overall performance, namely the F1 score, exhibited significant

progress, which highlighted an optimized balance between precision and recall in our segmentation process. In conclusion, the performance of the modified FCMs approach substantiates its superiority over classical methods and showcases its potential in refining the overall image segmentation process.

Figure 1 outlines the workflow for our enhanced Brain MRI Image Segmentation process utilizing the integration of the proposed FCM with GA. The process begins with image preprocessing that includes noise reduction and normalization. Following this, FCM algorithm is optimized by GA for parameter tuning based on the F1-Score, aimed at improving segmentation precision. The workflow culminates with the pixel-based classification and evaluation, where performance metrics are calculated from the confusion matrix to assess the efficacy of the segmentation. This streamlined approach demonstrates the model's capability for sophisticated MR image analysis.

6.3 Results

In this section, we delve into the results obtained from our investigation into the segmentation of brain MRI images using our proposed algorithm. By analyzing both the geometrical interpretations and the quantitative metrics detailed in Tables 7, 8, 9, 10, we aim to provide a

comprehensive overview of the algorithm's effectiveness. This not only encompasses its accuracy in pinpointing tumor regions within the brain but also its capacity to differentiate between tumorous and non-tumorous tissues. Such detailed scrutiny is pivotal for assessing the practical utility of our algorithm in clinical settings, where precision and reliability are paramount.

The geometrical interpretation and quantitative results of the metrics presented in Tables 7, 8, 9, 10 provide a deeper understanding of our algorithm's performance in brain MRI image segmentation. These interpretations help illustrate not only the algorithm's accuracy in identifying tumor regions but also its effectiveness in distinguishing between tumor and non-tumor areas.

- **F1 score:** Geometrically, the F1 Score represents the harmonic mean of precision and recall, illustrating the balance between the algorithm's accuracy in identifying true tumor pixels (true positives) and its ability to minimize false positives and false negatives. This metric effectively quantifies the overlap between the algorithm-identified tumor areas and the actual tumor areas, highlighting the segmentation's precision and reliability.
- **Precision:** Precision measures the ratio of correctly identified tumor pixels to all pixels identified as tumors by the algorithm. Geometrically, this indicates the extent to which the segmented tumor area correctly falls within the actual tumor boundaries. A high precision score signifies that the majority of the pixels labeled as tumor by the algorithm are true tumor pixels, minimizing over-segmentation.
- **Sensitivity (recall):** Sensitivity, or recall, reflects the proportion of actual tumor pixels that have been correctly identified as such by the algorithm. Geometrically, this metric assesses how much of the actual tumor area is captured by the algorithm's segmentation. High sensitivity indicates that the algorithm effectively identifies tumor areas without missing significant portions, which is crucial for accurate medical diagnoses.
- **Specificity:** Specificity measures the proportion of non-tumor pixels correctly identified as non-tumor by the algorithm. Geometrically, this metric evaluates how well the algorithm can delineate non-tumor areas,

ensuring that non-tumor tissues are not incorrectly classified as tumors. High specificity indicates effective exclusion of non-tumor areas from the segmented tumor region, reducing false positives.

Incorporating these geometrical interpretations alongside the numerical results in Tables 7, 8, 9, 10 enriches our discussion on the algorithm's segmentation performance. It provides a clearer visual and mathematical explanation of how well the proposed algorithm can differentiate between tumor and non-tumor regions, further substantiating the effectiveness of our method in medical image segmentation tasks (Figs. 2, 3, 4, 5).

In this study, the innovative implementation of FCM algorithm showcases the power of integrating fuzzy set theory, fuzzy metrics, and Sugeno negation principles. Descriptive analyses clearly illustrate that the proposed algorithm achieves higher accuracy, precision, and recall rates compared to traditional FCM methods. Specifically, during experiments conducted on the BraTS2018 dataset, FCM + M1 + Sugeno, FCM + M2 + Sugeno, and FCM + M3 + Sugeno algorithms obtained F1 scores of 86.99%, 89.75%, and 89.62%, respectively. These figures represent a notable improvement over the 85.30% F1 score achieved by the conventional FCM method.

The optimization process unfolds across several generations, each refining the parameter set based on a fitness function that evaluates segmentation efficacy. Specifically, F1 score was utilized as the fitness function due to its balanced assessment of precision and recall, crucial for gauging the quality of image segmentation. This evolutionary process facilitates continuous improvement, with selection, crossover, and mutation operations generating progressively optimized parameter sets. Such a methodological approach ensures that the GA efficiently explores the parameter space, avoiding local optima and steadily improving segmentation outcomes. Observations from the application of GA demonstrate substantial enhancements in segmentation metrics, including accuracy, precision, and F1 scores, which are indicative of the algorithm's capacity to fine-tune parameters effectively across generations. The iterative refinement process highlights GA's role in mitigating overfitting risks while optimizing for global solutions, showcasing its critical contribution to advancing the performance of FCM algorithm in complex imaging

Table 7 Performance metrics results on Brats18_2013_2_1_flair image

Method	ACC	SP	SE	PR	F1	IoU
FCM + M1 + SUGENO	99.25%	99.35%	95.66%	79.76%	86.99%	76.98%
FCM + M2 + SUGENO	99.37%	99.59%	92.25%	87.38%	89.75%	81.40%
FCM + M3 + SUGENO	99.36%	99.58%	92.23%	87.15%	89.62%	81.20%
FCM	99.18%	99.23%	96.95%	76.15%	85.30%	74.37%

Table 8 Performance metrics results on brats18_2013_10_1_flair image

Method	ACC	SP	SE	PR	F1	IoU
FCM + M1 + SUGENO	99.70%	99.94%	81.48%	95.35%	87.87%	78.37%
FCM + M2 + SUGENO	99.69%	99.90%	83.35%	91.48%	87.23%	77.35%
FCM + M3 + SUGENO	99.35%	99.99%	63.69%	99.38%	77.63%	63.43%
FCM	99.70%	99.92%	82.33%	93.80%	87.69%	78.09%

Table 9 Performance metrics results on Brats18_2013_7_1_flair image

Method	ACC	SP	SE	PR	F1	IoU
FCM + M1 + SUGENO	99.25%	99.21%	99.92%	85.52%	92.16%	85.46%
FCM + M2 + SUGENO	99.17%	99.13%	99.96%	84.01%	91.29%	83.98%
FCM + M3 + SUGENO	99.58%	99.61%	99.03%	92.93%	95.88%	92.09%
FCM	99.25%	99.21%	99.92%	85.52%	92.16%	85.46%

Table 10 Performance metrics results on brats18_2013_9_1_flair image

Method	ACC	SP	SE	PR	F1	IoU
FCM + M1 + SUGENO	99.21%	99.47%	83.96%	73.47%	78.36%	64.43%
FCM + M2 + SUGENO	99.25%	99.42%	88.11%	70.68%	78.44%	64.53%
FCM + M3 + SUGENO	98.86%	99.62%	67.13%	80.84%	73.35%	57.92%
FCM	98.86%	99.62%	67.13%	80.84%	73.35%	57.92%

scenarios. These improvements underscore the pivotal impact of GA in the optimization of learning models, particularly in the context of medical image segmentation, where precision is paramount.

Incorporating boxplot visualization in Fig. 6 provides a comprehensive and comparative analysis of the performance of various FCM algorithm configurations, namely Standard FCM, FCM + M1 + Sugeno, FCM + M2 + Sugeno, and FCM + M3 + Sugeno, as evidenced by their F1 scores. This visualization serves as a pivotal element in our results section, illustrating not only the distribution and central tendency of the performance metrics for each configuration but also highlighting potential outliers that might indicate variability in performance under certain conditions. One of the most notable observations from this analysis is the consistently higher F1 scores achieved by FCM + M3 + Sugeno configuration, suggesting its enhanced capability in handling image segmentation tasks, particularly in scenarios that may pose specific challenges. The superior performance of FCM + M3 + Sugeno underscores the effectiveness of integrating Sugeno negation and optimization techniques in refining the segmentation process, offering a promising avenue for future research and application. Furthermore, the width of the boxplots and the identification of outliers within these visual representations offer valuable insights into the consistency and reliability of each method across varied datasets. Narrower boxplots denote a higher level of consistency in the results produced by a method, underscoring

its robustness. Conversely, the presence of outliers may signal that the method could exhibit unexpected performance deviations under certain conditions, warranting further investigation to understand and mitigate such occurrences. By providing a clear and intuitive visual comparison of FCM configurations, this boxplot visualization significantly enriches our manuscript. It not only facilitates a deeper understanding and assessment of each configuration's performance but also highlights the strengths and areas for improvement of our proposed modifications to FCM algorithm. This comprehensive analysis and visualization strategy thus plays a crucial role in illustrating the potential impact of our work on advancing the field of image segmentation and guiding future efforts toward the development of more accurate and adaptable segmentation algorithms.

Furthermore, SE and SP scores highlighted by the descriptive analysis results present significant findings. FCM + M1 + Sugeno method exhibited remarkable performance with a sensitivity rate of 95.66% and a specificity rate of 99.35%, while FCM + M2 + Sugeno and FCM + M3 + Sugeno methods achieved sensitivity rates of 92.25% and 92.23%, and specificity rates of 99.59% and 99.58%, respectively. These outcomes demonstrate the superiority of the proposed methodologies in differentiating between tumor and non-tumor regions compared to the traditional FCM algorithm. These descriptive analyses validate the precision and effectiveness of the proposed FCM algorithms in solving complex problems like medical

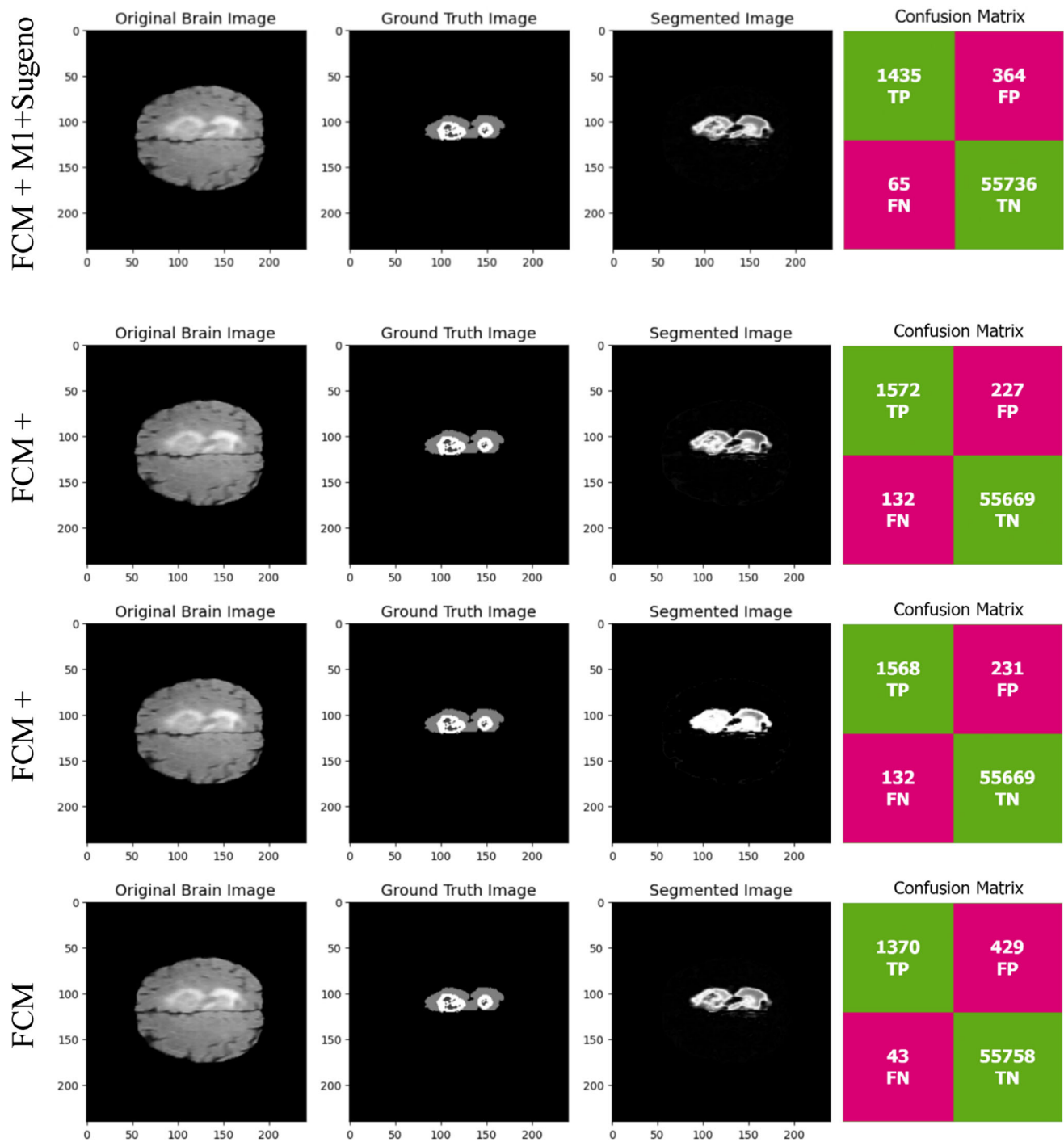


Fig. 2 Visualization of segmentation results on Brats18_2013_2_1_flair image

image segmentation, characterized by uncertainty, noise, and ambiguity. The evident improvement in the algorithms' performance is a direct result of the strategic use of fuzzy set theory and the optimization of parameters through GA. This work highlights the potential of integrating fuzzy logic and GA to enhance the performance of

machine learning models and establishes a solid foundation for future research in this field.

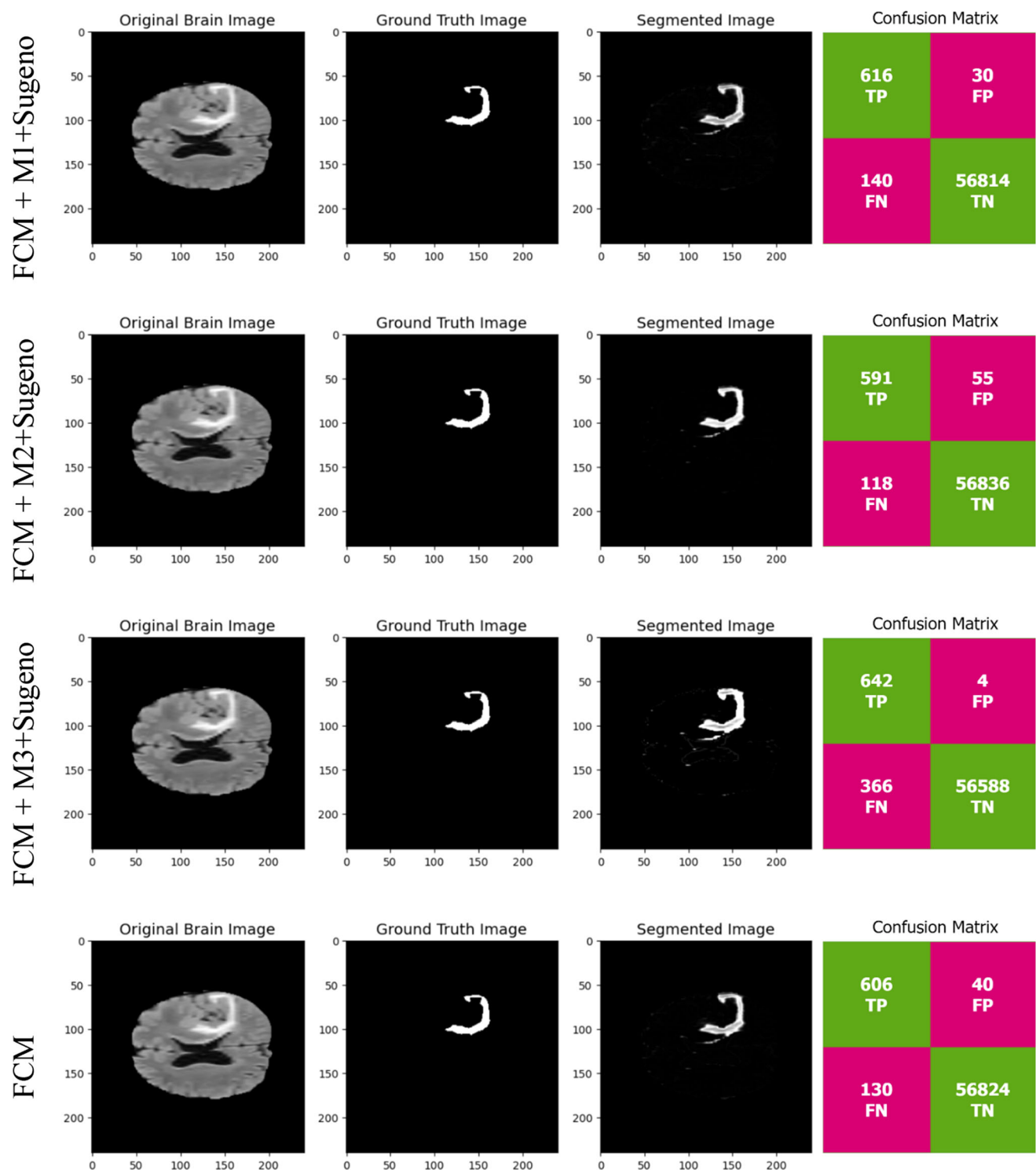


Fig. 3 Visualization of segmentation results on Brats18_2013_10_1_flair image

7 Conclusion and discussion

This study represents a significant leap forward in the application of fuzzy logic within the realm of machine learning, particularly in the segmentation of brain MRI

images. By integrating fuzzy set theory, fuzzy metrics, and the Sugeno negation operator into FCM algorithm and optimizing it through GA, we have unveiled a methodology that not only transcends traditional machine learning constraints but also exhibits unparalleled adaptability and

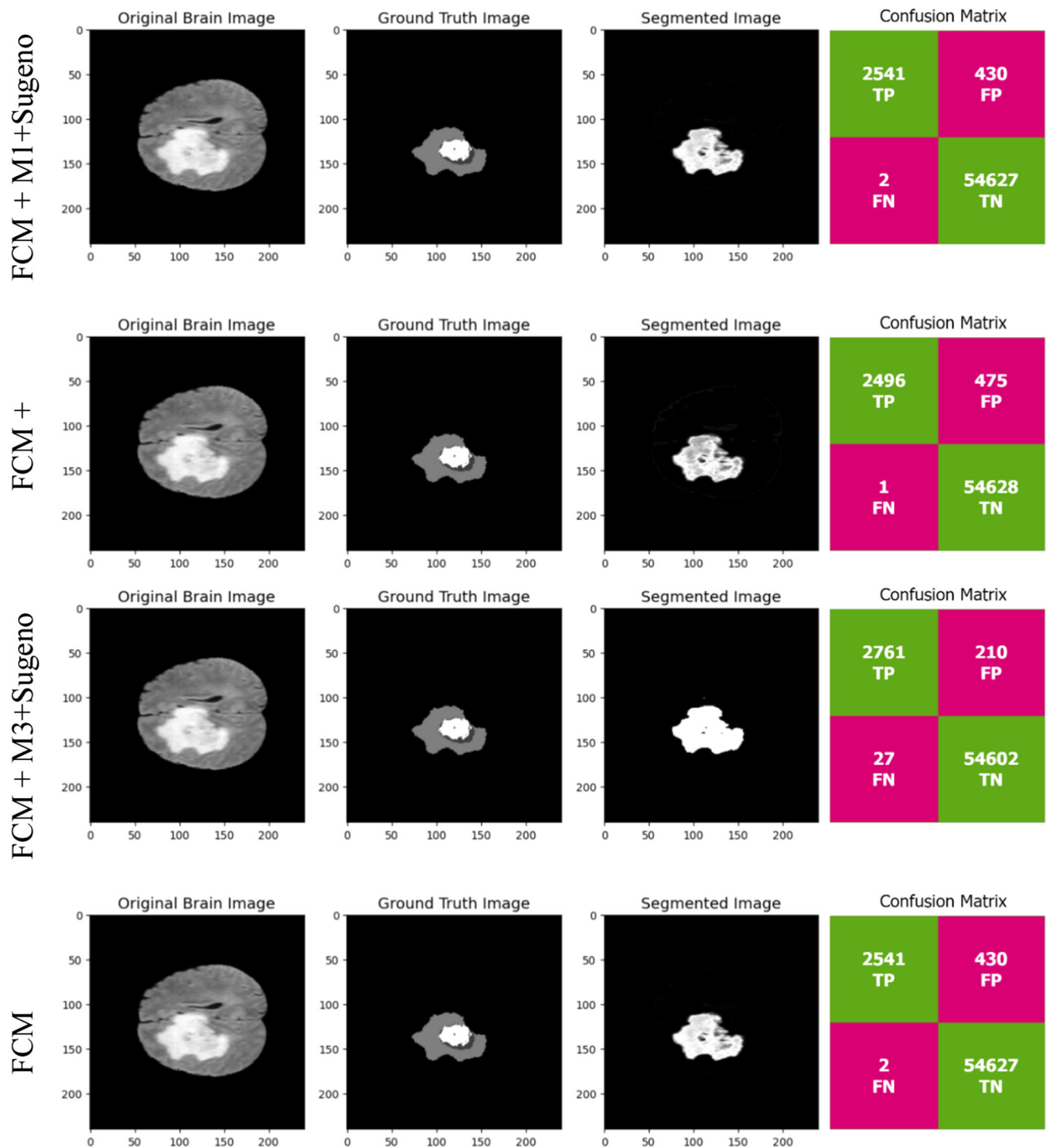


Fig. 4 Visualization of segmentation results on Brats18_2013_7_1_flair image

precision in the face of data ambiguity, noise, and imprecision. Our research stands at the forefront of innovation in machine learning and image segmentation through its comprehensive and multifaceted contributions, which are elaborated as follows:

- *Revitalized FCM algorithm:* We have not merely adjusted but fundamentally transformed FCM algorithm. This transformation involves a pioneering integration of fuzzy set theory, fuzzy metrics, and the Sugeno negation operator. By doing so, our approach bridges the gap between traditional segmentation methods and the need for more adaptive, nuanced

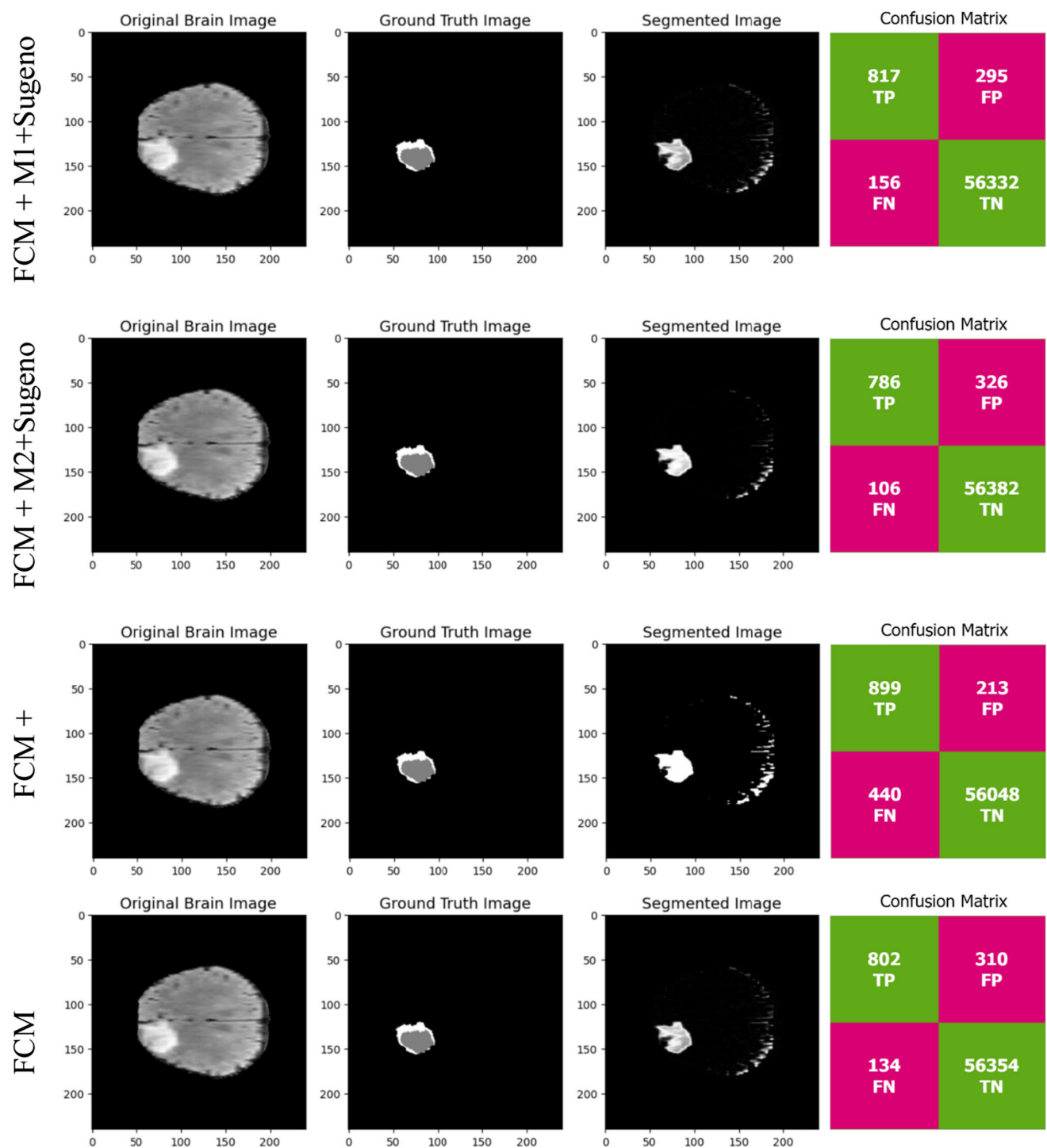


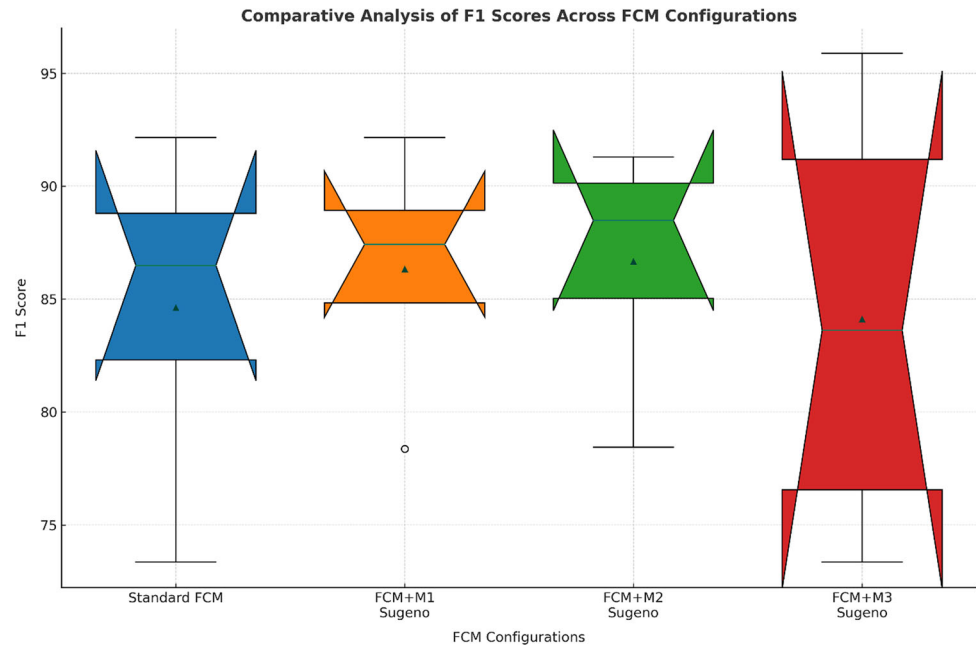
Fig. 5 Visualization of segmentation results on brats18_2013_9_1_flair image

solutions in the face of data that is inherently ambiguous, noisy, and imprecise. This represents a paradigm shift in how algorithms adapt to and interpret complex datasets.

- *Advanced clustering mechanism through fuzzy logic:* Our study elevates the clustering mechanism by leveraging the nuanced capabilities of fuzzy logic. The

introduction of fuzzy metrics to assess proximity and the innovative use of the Sugeno negation operator to interpret these metrics as distances offer a more sophisticated, flexible clustering process. This approach allows for a dynamic adjustment to the inherent vagueness and overlaps found in real-world data,

Fig. 6 F1 scores comparison of FCM configurations



particularly in medical imaging, where such characteristics are prevalent.

- *Optimization with GA:* The strategic use of GA to optimize the parameters of the fuzzy metric and negation operators marks a significant advancement. This optimization ensures that the algorithm not only performs with maximal efficiency but also retains its adaptability across different datasets and segmentation challenges. It highlights our contribution in applying evolutionary computation techniques to fine-tune the algorithm, enhancing its sensitivity and specificity in segmenting MRI images.
- *Superior performance in MRI image segmentation:* A key contribution of our research is the demonstrated superior performance of the modified FCM algorithm in the segmentation of brain MRI images. Through rigorous experimentation and comparison with traditional methods, our approach has shown significant improvements in accuracy, precision, recall, and overall segmentation quality. This success underscores the potential of our methodology to improve diagnostic processes by providing more reliable and detailed image analyses.
- *Framework for future research in fuzzy-based machine learning:* Beyond its immediate applications, our study provides a robust framework for future research into fuzzy-based machine learning models. By showcasing the effectiveness of integrating fuzzy logic with GAOptimization, we pave the way for further explorations into other complex data analysis tasks beyond MRI segmentation. Our research invites the academic community to build upon our findings, explore new

applications, and continue advancing the boundaries of what machine learning algorithms can achieve.

- *Advantages and limitations:* The advantages of our approach include increased sensitivity to the nuances of data, resulting in higher accuracy and precision in segmentation tasks. Moreover, the adaptability of the algorithm allows for its application across various complex datasets, showcasing its robustness. However, a notable limitation is the increased computational demand, primarily due to the GA's optimization process. This aspect could potentially hinder the algorithm's applicability in real-time scenarios or when dealing with extensive datasets.

In conclusion, by ingeniously amalgamating fuzzy set theory, fuzzy metrics, and the Sugeno negation operator with GA, our study carves a novel pathway for the enhancement of machine learning algorithms for complex data segmentation tasks. The proposed methodology not only redefines FCM algorithm within an advanced fuzzy logic framework but also paves the way for future research, promising substantial advancements in the machine learning and data analysis domains.

8 Future works

Future research should focus on developing methods that reduce computational demands and enhance the generalizability of the algorithm. This would enable broader applications across larger datasets and more complex

problem domains. Looking ahead, several avenues for further research emerge:

- *Algorithm efficiency*: exploring techniques to reduce computational demands, such as parallel processing or more efficient optimization algorithms, will be crucial. This could widen the algorithm's applicability, making it feasible for larger datasets and real-time analysis.
- *Broader applications*: extending the use of our enhanced FCM algorithm beyond MRI image segmentation to other domains, such as natural language processing or environmental monitoring, carotid atherosclerotic plaque etc. [52, 53] could prove beneficial. This would test the algorithm's versatility and adaptability to various data types.
- *Development of fuzzy metrics*: crafting new fuzzy metrics tailored to specific segmentation challenges could further refine the algorithm, enhancing its precision and flexibility. This bespoke approach would allow for nuanced segmentation tasks to be conducted with even greater accuracy.

Author contributions Fatih Kutlu: conceptualization; formal analysis; methodology; resources; software; supervision; writing—original draft; writing—review and editing. İbrahim Ayaz: data curation; methodology; resources; software; validation; visualization; writing—original draft; writing—review and editing. Harish Garg: conceptualization; formal analysis; methodology; resources; software; supervision; writing—original draft; writing—review and editing.

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). There is no funding provider.

Data availability statement The dataset utilized in this study is sourced from BraTS2018, facilitated by the Section for Biomedical Image Analysis (SBIA) of the University of Pennsylvania. This dataset is publicly accessible at <https://www.med.upenn.edu/sbia/brats2018/data.html>. All the data used in our research is duly cited in the relevant sections of this document. The usage of this dataset includes detailed information and guidelines for other researchers in this field, and our study has been conducted in accordance with these directives.

Declarations

Conflict of interest The authors declare no competing interests.

Ethics approval I hereby declare that this study has been scrutinized and approved from an ethical standpoint. Throughout the research process, no ethical issues or conflicts have arisen. Furthermore, during the literature review, data collection, and analysis stages, complete adherence to principles of impartiality and accuracy has been maintained. All data has been obtained reliably and reported accurately. This statement is presented to confirm the full compliance of the research process with ethical standards. The usage rules of the open-access data set used in the study were followed.

Consent to participate Informed consent was waived by ethics groups.

Consent for publication Informed consent was waived by ethics groups.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Zadeh LA (1965) Fuzzy sets. *Inf Control* 8:338–353. [https://doi.org/10.1016/S0019-9958\(65\)90241-X](https://doi.org/10.1016/S0019-9958(65)90241-X)
2. Couso I, Borgelt C, Hullermeier E, Kruse R (2019) Fuzzy sets in data analysis: From statistical foundations to machine learning. *IEEE Comput Intell Mag* 14:31–44. <https://doi.org/10.1109/MCI.2018.2881642>
3. Dunn JC (1973) A fuzzy relative of the ISODATA process and its use in detecting compact well-separated clusters. *J Cybernet* 3:32–57. <https://doi.org/10.1080/01969727308546046>
4. Bezdek JC, Ehrlich R, Full W (1984) FCM: the fuzzy c-means clustering algorithm. *Comput Geosci* 10:191–203. [https://doi.org/10.1016/0098-3004\(84\)90020-7](https://doi.org/10.1016/0098-3004(84)90020-7)
5. Ghosh S, Kumar S (2013) Comparative analysis of K-means and fuzzy C-means algorithms. *Int J Adv Comput Sci Appl* <https://doi.org/10.14569/ijacsa.2013.040406>
6. Gao Y, Wang S, Liu S (2009) Automatic clustering based on GA-FCM for pattern recognition. In: *ISCID 2009 - 2009 international symposium on computational intelligence and design*. pp 146–149
7. Tang Y, Ren F, Pedrycz W (2020) Fuzzy C-means clustering through SSIM and patch for image segmentation. *Appl Soft Comput* 87:105928. <https://doi.org/10.1016/J.ASOC.2019.105928>
8. Özyurt F, Sert E, Avcı D (2020) An expert system for brain tumor detection: Fuzzy C-means with super resolution and whale convolutional neural network with extreme learning machine. *Med Hypotheses* 134:109433. <https://doi.org/10.1016/J.MEHY.2019.109433>
9. Cheng-Bing L, Xi-hao M (2021) Array sensors online pattern recognition based on FCM and ANFIS. *Int J Comput Appl* 43:352–359. <https://doi.org/10.1080/1206212X.2018.1550167>
10. Hua L, Gu Y, Gu X et al (2021) A novel brain MRI image segmentation method using an improved multi-view fuzzy c-means clustering algorithm. *Front Neurosci*. <https://doi.org/10.3389/fnins.2021.662674>
11. Tongbram S, Shimray BA, Singh LS, Dhanachandra N (2021) A novel image segmentation approach using fcm and whale optimization algorithm. *J Ambient Intell Humaniz Comput* 1:1–15. <https://doi.org/10.1007/S12652-020-02762-W/TABLES/11>
12. Shi Z, Wu D, Guo C et al (2021) FCM-RDPa: TSK fuzzy regression model construction using fuzzy C-means clustering,

- regularization, Droprule, and Powerball Adabelief. *Inf Sci (N Y)* 574:490–504. <https://doi.org/10.1016/j.ins.2021.05.084>
13. Ayyappa Y, Neelakanteswara P, Bekkanti A, et al (2021) automatic face mask recognition system with FCM and BPNN. In: *Proceedings - 5th international conference on computing methodologies and communication, ICCMC 2021*. Institute of Electrical and Electronics Engineers Inc., pp 1134–1137
 14. Zhang X, Ning Y, Li X, Zhang C (2021) Anti-noise FCM image segmentation method based on quadratic polynomial. *Signal Process* 178:107767. <https://doi.org/10.1016/j.sigpro.2020.107767>
 15. Murfi H, Rosaline N, Hariadi N (2022) Deep autoencoder-based fuzzy c-means for topic detection. *Array* 13:100124. <https://doi.org/10.1016/J.ARRAY.2021.100124>
 16. Meng F (2021) Safety warning model of coal face based on fcm fuzzy clustering and ga-bp neural network. *Symmetry (Basel)*. <https://doi.org/10.3390/sym13061082>
 17. Karczmarek P, Kiersztyn A, Pedrycz W, Czerwiński D (2021) Fuzzy C-Means-based Isolation Forest. *Appl Soft Comput* 106:107354. <https://doi.org/10.1016/J.ASOC.2021.107354>
 18. Yu JW, Kim JS, Li X et al (2022) Water quality forecasting based on data decomposition, fuzzy clustering and deep learning neural network. *Environ Pollut* 303:119136. <https://doi.org/10.1016/J.ENVPOL.2022.119136>
 19. Gosain A, Dahiya S (2016) Performance analysis of various fuzzy clustering algorithms: a review. *Procedia Comput Sci* 79:100–111. <https://doi.org/10.1016/J.PROCS.2016.03.014>
 20. Ruspini EH, Bezdek JC, Keller JM (2019) Fuzzy clustering: a historical perspective. *IEEE Comput Intell Mag* 14:45–55. <https://doi.org/10.1109/MCI.2018.2881643>
 21. Kramosil I, Michalek J (1975) Fuzzy metrics and statistical metric spaces. *Kybernetika* 11:336–344
 22. George A, Veeramani P (1994) On some results in fuzzy metric spaces. *Fuzzy Sets Syst* 64:395–399. [https://doi.org/10.1016/0165-0114\(94\)90162-7](https://doi.org/10.1016/0165-0114(94)90162-7)
 23. George A, Veeramani P (1997) Short communication: on some results of analysis for fuzzy metric spaces. *Fuzzy Sets Syst* 90:349–353
 24. Park JH (2004) Intuitionistic fuzzy metric spaces. *Chaos Solitons Fractals* 22:1039–1046. <https://doi.org/10.1016/j.chaos.2004.02.051>
 25. Gregori V, Morillas S, Sapena A (2011) Examples of fuzzy metrics and applications. *Fuzzy Sets Syst* 170:95–111. <https://doi.org/10.1016/j.fss.2010.10.019>
 26. Kutlu F, Tuncdemir K (2021) Temporal intuitionistic fuzzy metric spaces. *Maejo Int J Sci Technol* 15:209–221
 27. Ralević NM, Karaklić D, Pištinjat N (2019) Fuzzy metric and its applications in removing the image noise. *Soft comput* 23:12049–12061. <https://doi.org/10.1007/S00500-019-03762-5/METRICS>
 28. Ralević N, Paunović M (2021) Applications of the fuzzy metrics in image denoising and segmentation. *Tehnički Vjesnik* 28:819–826
 29. Ralević NMR, Delicdelic M, Nedovic LN (2022) Aggregation of fuzzy metrics and its application in image segmentation. *Iran J Fuzzy Syst* 19:19–37
 30. Pham DL, Prince JL (1999) Adaptive fuzzy segmentation of magnetic resonance images. *IEEE Trans Med Imaging* 18:737–752. <https://doi.org/10.1109/42.802752>
 31. Siyal MY, Yu L (2005) An intelligent modified fuzzy c-means based algorithm for bias estimation and segmentation of brain MRI. *Pattern Recognit Lett* 26:2052–2062. <https://doi.org/10.1016/J.PATREC.2005.03.019>
 32. Singh C, Bala A (2018) A DCT-based local and non-local fuzzy C-means algorithm for segmentation of brain magnetic resonance images. *Appl Soft Comput* 68:447–457. <https://doi.org/10.1016/J.ASOC.2018.03.054>
 33. Bai X, Zhang Y, Liu H, Chen Z (2019) Similarity measure-based possibilistic FCM with label information for brain MRI segmentation. *IEEE Trans Cybern* 49:2618–2630. <https://doi.org/10.1109/TCYB.2018.2830977>
 34. Huang H, Meng F, Zhou S et al (2019) Brain image segmentation based on FCM clustering algorithm and rough set. *IEEE Access* 7:12386–12396. <https://doi.org/10.1109/ACCESS.2019.2893063>
 35. Liu Z, Bai X, Liu H, Zhang Y (2020) Multiple-surface-approximation-based FCM with interval memberships for bias correction and segmentation of brain MRI. *IEEE Trans Fuzzy Syst* 28:2093–2106. <https://doi.org/10.1109/TFUZZ.2019.2930478>
 36. Valsalan P, Sriramakrishnan P, Sridhar S et al (2020) Knowledge based fuzzy c-means method for rapid brain tissues segmentation of magnetic resonance imaging scans with CUDA enabled GPU machine. *J Ambient Intell Humaniz Comput*. <https://doi.org/10.1007/S12652-020-02132-6/METRICS>
 37. Tavakoli-Zaniani M, Sedighi-Maman Z, Fazel Zarandi MH (2021) Segmentation of white matter, grey matter and cerebrospinal fluid from brain MR images using a modified FCM based on double estimation. *Biomed Signal Process Control* 68:102615. <https://doi.org/10.1016/J.BSPC.2021.102615>
 38. Neves-Jr F, Arruda LVR, Mendonça M (2009) A combined FCM-GA approach to supervise industrial process. *IFAC Proc Vol* 42:1144–1149. <https://doi.org/10.3182/20090630-4-ES-2003.00188>
 39. Chen T-C, You P-S, Wu C-H, Lin S-L (2014) Using FCM based hybrid computational approach for diseases diagnosis in traditional chinese medicine. *Int J Mach Learn Comput* 4:389–393. <https://doi.org/10.7763/IJMLC.2014.V4.442>
 40. Ghazanfari M, Alizadeh S, Fathian M, Koulouriotis DE (2007) Comparing simulated annealing and genetic algorithm in learning FCM. *Appl Math Comput* 192:56–68. <https://doi.org/10.1016/J.AMC.2007.02.144>
 41. Wang LW, Zhu YQ, Pan YF (2005) FCM algorithm and index CS for the signal sorting of radiant points. In: *2005 international conference on machine learning and cybernetics, ICMLC 2005*, pp 4415–4419. <https://doi.org/10.1109/ICMLC.2005.1527716>
 42. Bede B (2013) *Mathematics of fuzzy sets and fuzzy logic*. Springer, Berlin
 43. Holland JH (1992) Genetic algorithms. *Sci Am* 267:66–73
 44. Li S, Li D (2021) Genetic algorithms. In: *Springer Series in Materials Science*. pp 115–131
 45. Katoch S, Chauhan SS, Kumar V (2021) A review on genetic algorithm: past, present, and future. *Multimed Tools Appl* 80:8091–8126. <https://doi.org/10.1007/s11042-020-10139-6>
 46. Zeebaree DQ, Haron H, Abdulazeez AM, Zeebaree SRM (2017) Combination of k-means clustering with genetic algorithm: a review. *Int J Appl Eng Res* 12:14238–14245
 47. Ding Y, Fu X (2016) Kernel-based fuzzy c-means clustering algorithm based on genetic algorithm. *Neurocomputing* 188:233–238. <https://doi.org/10.1016/j.neucom.2015.01.106>
 48. Li H, Wang F, Li H (2019) Integrating expert knowledge for Bayesian network structure learning based on intuitionistic fuzzy set and genetic algorithm. *Intell Data Anal* 23:41–56. <https://doi.org/10.3233/IDA-183877>
 49. Bakas S, Reyes M, Jakab A, et al (2018) Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the BRATS challenge. *Sandra Gonzalez-Vill* 124:
 50. Menze BH, Jakab A, Bauer S et al (2015) The multimodal brain tumor image segmentation benchmark (BRATS). *IEEE Trans Med Imaging* 34:1993–2024. <https://doi.org/10.1109/TMI.2014.2377694>

51. Bakas S, Akbari H, Sotiras A et al (2017) Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features. *Sci Data*. <https://doi.org/10.1038/SDATA.2017.117>
52. Huo R, Liu Y, Xu H, Li J, Xin R, Xing Z, Deng S, Wang T, Yuan H, Zhao X (2022) Associations between carotid atherosclerotic plaque characteristics determined by magnetic resonance imaging and improvement of cognition in patients undergoing carotid endarterectomy. *Quant Imaging Med Surg* 12(5):2891–2903
53. Han M, He W, He Z, Yan X, Fang X (2022) Anatomical characteristics affecting the surgical approach of oblique lateral lumbar interbody fusion: an MR-based observational study. *J Orthop Surg Res* 17(1):426. <https://doi.org/10.1186/s13018-022-03322-y>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.